

Paying for Performance

Are Swedish Active Funds Worth the Extra Cost?

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Abstract

This study investigates whether Swedish active funds provide value to investors relative to their fund fees. Using Swedish fund data with 12 passive and 25 active equity funds from 2015 to 2025, the performance measure alpha was extracted using a Jensen single-factor and a Fama-French three-factor model. By applying a bootstrap technique, the funds with the highest alphas were then evaluated to see if active fund managers picked stocks well enough to more than cover their fund fees while controlling for luck. Two-sample t-tests were also performed on total return and risk-adjusted performance (Sharpe ratio) to analyze differences. The results show that actively managed funds do not on average, outperform passive funds. Passive funds delivered more stable and consistent results while active funds exhibited a larger variation in return, Sharpe ratio, and alpha. Only one fund had a significant alpha from the bootstrap analysis, indicating that these outcomes are rare. This specific result was based on limited data (2015 to 2019), affecting its reliability. Overall, the results showed that while active funds do manage to generate higher returns, fees erode these returns, making passive funds the better option on average. In conclusion, there is no evidence supporting that active funds are a better choice, especially since only a minority of active funds manage to outperform after fees, while controlling for luck.

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1 Introduction

In Sweden 2024, the total fund assets amounted to 8,355 billion Swedish kronor, and the net inflow to funds was 208 billion Swedish kronor (Fondbolagens förening, 2025). Swedes are known for being diligent retail investors, more than seven out of ten Swedes hold mutual fund investments (Fondbolagens förening, n.d.). Sweden has one of the highest rates of mutual fund ownership in the world, with a large proportion of the population investing in funds (Tuvhag, 2023). Investing in funds mostly involves paying a management fee to the fund manager responsible for handling the portfolio. Typically, this fee is much lower for passively managed funds than for actively managed ones (Söderberg&Partners, 2014). The widespread participation in fund investing raises important questions. The central question is whether the higher fees charged by actively managed funds are justified by better performance. To explore this, this thesis compares active and passive funds based on both total and risk-adjusted returns. For the active funds that managed to outperform the market after fees, a bootstrap evaluation is conducted to determine whether such excess returns can be attributed to genuine managerial skill or are simply the result of random variation. In doing so, this thesis aims to contribute to the ongoing discussion about cost-efficiency and return in fund management.

Choosing the right fund can have a significant impact on an individual's financial future. Over time, even small differences in annual returns can compound into substantial amounts, ultimately affecting one's lifestyle, the ability to travel, housing choices, and, perhaps most critically, the quality of one's retirement. For many Swedes, mutual funds are not just an abstract financial instrument but a core component of their long-term wealth-building strategy (Fondbolagens förening, n.d.). In light of this, understanding the performance of different types of funds becomes not only relevant but crucial. With thousands of options available, each with different categories, strategies, risk levels, and fee structures, choosing the right fund can be overwhelming for the average investor. Although investing in mutual funds is relatively straightforward today, understanding the wide range of available options may demand time and financial literacy. As a result, a simpler approach could be choosing a passively managed fund designed to mirror a market index. These funds typically come with lower management fees and aim to deliver returns in line with the broader market. In contrast, actively managed funds offer the promise of outperforming the market through skilled management and strategic decision-making, though this often comes at a higher cost (Söderberg&Partners, 2014). The question that will be investigated further in this thesis is:

Do actively managed funds in Sweden offer added value to retail investors compared to passive funds?

There have been several studies examining the differences in returns between actively and passively managed funds, with varying results. A key study in the debate on fund performance is an article by (Fama, 1970) which introduces the Efficient Market Hypothesis (EMH). EMH explains how the share price incorporates all available information, implying a fully rational market. According to this theory, the information is already reflected in the price, and it should therefore be extremely difficult for an investor to consistently beat the market. This article is relevant to this study since if this theory were correct, active funds should, on average, not be able to beat passive funds, and in the end, the two fund types should not differ on average. This would mean that in the long run, the extra fees for active funds would not be worth investing in. Although EMH is an important concept in economic history, several reports question its results.

A further important study by (Shiller, 1981) showed that stocks tend to be much more volatile than what

could be explained by available information, a result that questions whether the EMH is correct. In the article, Shiller compared historic stock prices with theoretical prices using the actual dividends that were paid out in the future, the so-called "ex-post rational prices". He found that volatility appeared to be up to thirteen times higher than his model predicted, which suggests that sentiment or speculative behavior affects the pricing movements. This theory challenges Fama and the Efficient Market Hypothesis and supports the idea of inefficiencies that active fund managers might be able to exploit to achieve higher returns, thereby making it highly relevant to this study. Furthermore in an article by (De Bondt and Thaler, 1985), a result shown was that stocks that performed badly often overperformed later, a pattern further challenging EMH. There is, therefore a lack of consensus regarding the Efficient Market Hypothesis.

A further question that is also debated when studying whether the higher fees across active funds are justified is whether the outperforming active funds achieve higher returns that cover more than their fund fees due to skill or luck. If the outperformance is primarily driven by randomness, active funds become difficult to justify since identifying those who manage to consistently cover more than the fund fees would be very hard. Several important studies have been made discussing this.

One article by (Fama and French, 2010) investigates whether high mutual fund returns could be explained by manager skill or if the results are more likely due to luck. A large sample of U.S equity funds was studied, and by using bootstrap simulation techniques, they studied whether differences in fund returns were statistically separable from random chance. The results found little evidence supporting that higher returns were a result of skill, suggesting instead that differences were a result of other factors, potentially including luck. A similar problem was studied in an article by (Kosowski et al., 2006). Here, performance was studied to determine if fund managers picked stocks well enough to more than cover their costs and if this is a result driven by skill or simply luck. A bootstrap approach was proposed to test this. Of the chosen 4200 funds, they found that only a small minority seemed to be a result of skill. This study will use a bootstrap approach inspired by this article to also test this.

This article contributes to earlier research by comparing Swedish equity funds that have been active for ten years. The aim is to provide insights for Swedish investors to find out whether the higher fees charged by actively managed funds in Sweden offer added value compared to passive funds, and whether such outperformance is due to managerial skill or pure chance.

In the next Section, the theoretical framework is presented, which provides the basis for this thesis. In Section 3, the methods used will be further discussed, along with how data was generated through the bootstrap evaluation. Section 4 consists of how the data were extracted, different performance measures used, and some descriptive statistics, while results from different hypothesis tests are presented in Section 5. Lastly, the results are concluded in Section 6.

2 Theory

In this section, an overview of key financial concepts, definitions, and methods is provided that are necessary to understand, as well as to compare different fund types. Each subsection introduces a core concept related to the financial market and explains its relevance to the study, along with the methods used to compare and analyze fund types.

2.1 Capital Asset Pricing Model

To evaluate the performance of actively managed funds compared to passive funds, the study will apply Jensen's alpha as a performance measure. Since this measure is derived from the Capital Asset Pricing model (CAPM), the theory for this will be presented first to understand the assumptions and logic behind the performance measure alpha, before it is then explained further.

CAPM was originally introduced by (Sharpe, 1964) and is used to describe the relationship between the risk of an asset and its expected return. It is based on the idea that investors require compensation for both the time value of money and for taking on more risk (Bodie, Kane, and Marcus, 2013). The model is based on assumptions in two main categories. The first category is related to investor behavior and their preferences regarding risk. It assumes that all investors evaluate the trade-off between risk and expected return in the same way. Investors are assumed to be equally informed, plan to invest over the same period, and form similar expectations about future returns. The second category is related to the financial market. CAPM assumes that the financial markets operate in an environment without friction, meaning trading is costless and untaxed. Moreover, investors are assumed to have the ability to borrow and lend at the same risk-free rate, as well as the ability to short sell. Short selling refers to selling assets that are not owned with the hope of buying them back later at a lower price. These assumptions contribute to equal opportunities for all investors. Together, they form the foundation of CAPM, leading to the idea that all investors will hold the same market portfolio, which consists of all available assets weighted in proportion to their market value (Bodie, Kane, and Marcus, 2013).

The formula for expected return according to CAPM is (Sharpe, 1964):

$$E(R_i) = R_F + \beta_{M,i}\,(E(R_M) - R_F) \tag{1}$$

where:

$E(R_i)$ is the expected return for the asset,

R_F is the risk free rate,

β_i is the assets sensitivity for market movements,

$E(R_M)$ is the expected return on market portfolio.

In this model, beta is the only variable that determines the expected return of an asset. It reflects how much the asset moves in relation to the overall market. A high beta value means the asset is more sensitive to overall market movements, indicating higher exposure to systematic risk (the type of risk that affects the entire market). According to this model, investors only expect higher returns while taking on more market risk (Bodie, Kane, and Marcus, 2013).

The risk-free rate used in the formula represents the return on an investment with no risk, usually measured on short-term securities like treasury bills. These are both safe and easy to sell since they are purchased at a lower price and paid back in full on a fixed date, providing a predictable return (Bodie, Kane, and Marcus, 2013). The risk-free rate used in this study will be the Swedish three-month treasury bills. Also, throughout this study R_M will refer to OMXSPI (OMX Stockholm Price Index) which refers to all shares listed on Nasdaq Stockholm. In this article, OMXSPI will simply be referred to as OMXS.

Several studies have been made regarding CAPM, including research that questions its assumptions and relevance. While CAPM provides a useful theoretical framework, it is based on simplified assumptions often not met in real markets. Several articles have criticized this. For example, the article "*A critique of the asset pricing theory's tests Part I: On past and potential testability of the theory*" argued that this meant CAPM could not be properly tested since some assumptions are not empirically verifiable (Roll, 1977). Furthermore, the article "*The Cross-Section of Expected Stock Returns*" also challenged the validity of the model by showing that factors such as size and book-to-market equity could explain average stock returns more efficiently than beta alone. They also found that beta on its own had limited explanatory power regarding expected return, contradicting CAPM (Fama and French, 1992). The critique of CAPM becomes particularly relevant since the study uses Jensen's alpha as a performance measure, which is directly based on CAPM. If the model does not hold in reality, the results of this report may be less reliable.

2.2 Fund and Fund Types

A fund consists of a variety of financial instruments like stocks, bonds, and other securities. Different kinds of funds allow for different focuses, such as sustainability and specific sectors (e.g, technology or energy). Purchasing a fund means acquiring shares of the fund, with each investor owning their proportional stake. Because a fund invests in various securities rather than just one, it generally tends to have lower risk, making it an attractive option for investors. This study focuses solely on equity funds that contain stocks. They are a popular form of investment option for investing in the stock market while also diversifying risk (Handelsbanken, n.d.). Funds can also be managed in two types of ways, active and passive (Söderberg&Partners, 2014).

A passive fund does not directly try to beat the market but instead follows a specific index, for example, S&P 500 on the US stock market or the OMXS on the Swedish stock market. The goal for passive funds is to try to replicate these indices and match their returns. Since passive fund management only tries to replicate a market index, the fees tend to be lower. Many passive funds, especially broad index funds, also tend to have better diversification, lowering the risk, while active funds tend to have fewer holdings, increasing the risk (Söderberg&Partners, 2014).

An active fund tries to consistently beat the market. This means that these funds have an opportunity for higher returns, while also being more flexible than passive funds that only try to replicate an index. Managers can also move away from downward trends and react to market changes, while passive funds will stick to following the index. Since these funds usually require more work and frequent adjustments and oversight, the fees tend to be higher. However, even if the goal for active funds is to beat the market, many fail to do this over time, especially when adjusting for the fees. The risk also tends to be higher since the performance depends on the decisions of the fund manager (Söderberg&Partners, 2014).

2.3 Two Sample t-test

The two-sample t-test will be used to study whether the mean for active funds before/after fees is greater than passive funds. This test is useful since it evaluates if there is a statistically significant difference between the means of the two samples (Triola, 2015). In this case, the two samples will consist of active and passive funds. The test will be evaluated both on total return and the Sharpe ratio.

The following hypothesis will be tested:

$$H_0 : \mu_{active} \leq \mu_{passive}$$
$$H_1 : \mu_{active} > \mu_{passive}$$

As seen in the hypothesis, the tests are conducted with the null hypothesis that the mean for the passive funds is less than or equal for the active funds, while the alternative hypothesis is that the mean for the active funds is greater than passive funds.

Before these tests can be performed, checks need to be made to see if the assumptions for the t-test are fulfilled (Triola, 2015). The t-test uses the following assumptions:

- σ_1 and σ_2 are unknown, and no assumptions that they are equal to each other is made
- Both samples are random
- The two samples are independent
- One or both of the following conditions are fulfilled:
  - The two sample sizes are both large, where n is bigger than 30
  - Both samples come from a normally distributed population

The assumption of unknown standard deviation is fulfilled, and that they are equal to each other is not assumed. Although the samples are not randomly drawn in this study, as Swedish equity funds with at least 10 years of available data were included, the aim was to analyze the entire relevant population within this scope. Therefore, statistical inference applies specifically to the selected funds rather than the full population of all possible funds. While the assumption of this random sampling is not fulfilled, t-tests are relatively robust to deviations from assumptions, particularly when other conditions such as approximate normality and independence are reasonably met (JMP, 2024). The last assumption is not fulfilled since the samples consist of 12 passive and 25 active funds; however, the normality assumption will be further evaluated to check if the data comes from a normal distribution. This can be done by inspecting QQ-plots along with the Shapiro Wilk test. The method is also reasonably robust against small deviations from normality in small samples, when no extreme outliers or skewness occur (Triola, 2015).

2.4 QQ-plots

To assess whether the variables in the dataset are approximately normally distributed to perform the two-sample t-test, quantile-quantile (QQ) plots will be utilized. A QQ-plot compares the observed against the expected quantiles from the normal distribution. If the data closely follows a normal distribution, the points in the QQ-plot will align along a straight diagonal line. Deviations from this line suggest departures from normality, such as skewness or heavy tails (Cryer and Chan, 2008). Using these QQ-plots allows for a visual and intuitive evaluation of the normality assumption, which is crucial for many statistical methods applied in this thesis, including certain parametric tests.

2.5 Shapiro Wilk test

To further study whether the data comes from a normal distribution, a Shapiro Wilk test can be performed. This test uses the following hypothesis (Shapiro and Wilk, 1965):

$$H_0 : \text{Data comes from a normal dsitribution}$$
$$H_1 : \text{Data does not come from a normal distribution}$$

This test is used since it helps to evaluate if data comes from a normal distribution or not by evaluating the p-value. If the null hypothesis cannot be rejected, it means that the data comes from a normal distribution can be assumed, which enables the performance of the two-sample t-test, given that all other assumptions are also met.

2.6 Single- and Three-Factor Models

Jensen's alpha is usually defined as the difference between actual and predicted returns based on CAPM (Bodie, Kane, and Marcus, 2013). This study will use the Jensen single-factor model and the Fama-French three-factor model to estimate alpha.

The Jensen single-factor model uses a regression to estimate each fund's alpha, and allows performance evaluation before and after fund fees, which helps to determine if any excess return beyond what is explained by market exposure remains, by analyzing the alpha. The three-factor model is also applied to further improve the accuracy of fund performance. The three-factor model is a further extension of the single-factor model, introduced by Fama and French (1993). This model includes two new factors, the first factor, SMB (small minus big), measures the additional return that small firms generate compared to large firms. The second factor, HML (high minus low), represents the difference in returns between companies with high and low book-to-market ratios. By also including SMB and HML, alpha estimation becomes more reliable by accounting for additional sources of systematic risk, allowing for a more accurate assessment of whether overperformance is a result of skill or pure random chance (Bodie, Kane, and Marcus, 2013).

2.7 Bootstrap

To further study whether the funds with the highest alphas after fees from the single- and three-factor model have outperformed by skill or chance, a bootstrap approach will be employed for this study. To estimate the uncertainty associated with a sample statistic, this thesis applies the bootstrap method, a resampling-based approach that does not rely on theoretical assumptions such as normality or the central limit theorem. Instead of drawing repeated samples from the population (as in traditional inference), the bootstrap method treats the observed sample as a proxy for the population itself and draws repeated samples with replacement from it. Formally, suppose an observed sample $x_1, x_2...x_n$ of size n, with interest in estimating a statistic $\hat{\theta}$. The bootstrap procedure proceeds as follows:

1. Resampling: Generate a random sample of size n with replacement from the original data. This new sample is called a bootstrap sample.
2. Statistic Calculation: Compute the value of the statistic $\hat{\theta}^b$ for each bootstrap sample
3. Repetition: Repeat steps 1 and 2 a large number of times to obtain a collection of bootstrap estimates $\hat{\theta}^b_1, \hat{\theta}^b_2...\hat{\theta}^b_B$

The resulting distribution of the bootstrap estimates approximates the sampling distribution of $\hat{\theta}$. This empirical distribution can then be used to assess the variability of $\hat{\theta}$, providing a flexible and robust inference framework, particularly when analytical solutions are intractable or assumptions of classical methods are violated. (Devore and Berk, 2012). The underlying null hypothesis in this study is that no fund manager was able to generate returns above the benchmark after adjusting for risk, i.e:

$$H_0 : \alpha_i \leq 0$$

$$H_1 : \alpha_i > 0$$

3 Method

This section outlines the methodology needed to compare fund performance and assess whether active funds provided added value. The section outlines how estimators were calculated along with test statistics, significance level, and sample sizes for the methods introduced in the theory. For the study, a significance level of 0.05 was used throughout the analysis since this is a commonly used and accepted standard in statistical reports. All tests were performed using the statistical software R.

First, the tests for normality were performed. QQ-plots were plotted in R to visually inspect the distributions for total return before and after fees, along with the Sharpe ratio. After this, the Shapiro Wilk test for normality were performed to check if the normality assumption holds. The test is performed in R with the following test statistic:

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \tag{2}$$

where:

a_i is the weight used to check how close the data is to a normal distribution,

x_i is the i:th value in the sample,

$\bar{x}$ is the sample mean.

After checking for normality, the two-sample t-test was performed. The two-sample t-test with the following test was performed on both total return and the Sharpe ratio, with the following test statistic:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (d_0)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \tag{3}$$

where:

$\bar{x}_1$ is the sample mean in group 1,

$\bar{x}_2$ is the sample mean in group 2 ,

d_0 is the hypothesized difference between two groups,

s_1^2 is the sample variance in group 1,

s_2^2 is the sample variance in group 2.

The sample size corresponds with the number of selected funds, 12 passive and 25 active funds, as the aim was to include the entire relevant and available population within the scope of Swedish equity funds for the 10 year period. Finally, results were evaluated based on whether the difference in mean return between the fund types is statistically significant.

After this, to further evaluate and compare the fund performance, Jensen's alpha was estimated through the single- and three-factor models. The Jensen single-factor model is defined as:

$$R_{i,t} = \alpha_i + \beta_{M,i} \cdot R_{M,t} + \varepsilon_{i,t} \tag{4}$$

where:

$R_{i,t}$ is the excess return of portfolio i during month t,

α_i is the measure of abnormal performance,

$R_{M,t}$ is the excess return on market portfolio during month t,

$\varepsilon_{i,t}$ is the residual of fund i during month t.

Excess return is simply how much more a portfolio earns than the risk-free rate (Bodie, Kane, and Marcus, 2013). This model allows the study to assess whether active funds generate large enough excess returns to provide added value or not by studying the alphas extracted by the model. After this, the three-factor model was also used to estimate alpha to improve the accuracy of the performance measure. The three-factor serves as an extension of the single-factor model by controlling for factors, small minus big (SMB) and high minus low (HML) (Bodie, Kane, and Marcus, 2013).

The Fama-French three-factor model is defined as:

$$R_{i,t} = \alpha_i + \beta_{M,i} \cdot R_{M,t} + \beta_{SMB,i} \cdot SMB_t + \beta_{HML,i} \cdot HML_t + \varepsilon_{i,t} \tag{5}$$

where:

SMB_t is the return difference between small and large firms,

HML_t is return difference between high and low book-to-market firms.

After running the factor models for each fund, the fund-specific alpha, beta, and residual estimates were obtained. The estimated residuals $\hat{\varepsilon}_{i,t}$ were then saved and drawn with replacement to simulate return series under the null hypothesis, imposing that $\alpha_i = 0$ for all funds. For each bootstrap iteration, synthetic returns were constructed using the estimated betas and resampled residuals to match the fund's risk profile for each model:

$$R^b_{i,t} = 0 + \hat{\beta}_{M,i} \cdot R^b_{M,t} + \hat{\varepsilon}^b_{i,t} \tag{6}$$

and

$$R^b_{i,t} = 0 + \hat{\beta}_{M,i} \cdot R^b_{M,t} + \hat{\beta}_{SMB,i} \cdot SMB^b_t + \hat{\beta}_{HML,i} \cdot HML^b_t + \hat{\varepsilon}^b_{i,t} \tag{7}$$

This procedure made it possible to approximate the sampling distribution of $\hat{\alpha}_i$ under the null hypothesis of no abnormal performance ($\alpha_i = 0$) for each fund. By repeating the regression for each fund, a distribution of bootstrap estimates $\hat{\alpha}_i^b$ was constructed. In each iteration, a new alpha estimate was calculated:

$$R_{i,t}^b = \hat{\alpha}_i^b + \hat{\beta}_{M,i}^b \cdot R_{M,t}^b + \hat{\varepsilon}_{i,t}^b \tag{8}$$

and

$$R_{i,t}^b = \hat{\alpha}_i^b + \hat{\beta}_{M,i}^b \cdot R_{M,t}^b + \hat{\beta}_{SMB,i}^b \cdot SMB_t^b + \hat{\beta}_{HML,i}^b \cdot HML_t^b + \hat{\varepsilon}_{i,t}^b \tag{9}$$

By comparing the original alpha to its corresponding empirical bootstrap distribution, the proportion of bootstrap estimates $\hat{\alpha}_i^b$ greater than the observed $\hat{\alpha}_i$ was interpreted as the p-value under the null hypothesis $\alpha_i \leq 0$ through:

$$\text{p-value} = \frac{1}{B} \sum_{b=1}^{B} \mathbb{I}(\hat{\alpha}^b \geq \hat{\alpha}) \tag{10}$$

where:

B is the number of bootstrap replications,

$\hat{\alpha}^b$ is the b-th bootstrap estimate of alpha,

$\hat{\alpha}$ is the observed alpha from the original sample,

$\mathbb{I}$ is the indicator function, which equals 1 if the condition inside is true, and 0 otherwise.

The observed performance (alpha) was assessed to determine if it could plausibly be attributed to luck alone. If the actual alpha fell in the extreme right tail of the bootstrap distribution, it was interpreted as an evidence of skill. The process was repeated 10,000 times for each fund. Although the original article by (Kosowski et al., 2006) used 1000 bootstrap iterations, 10,000 iterations is motivated since it resulted in a more stable sampling distribution.

4 Data

This section outlines the data processing steps, describes the measurement metrics used in the analysis, and presents relevant visualizations. By filtering funds in Morningstar, all funds with the "Morningstar category" Sweden and "registered" country Sweden were chosen. Monthly returns were examined from Refinitiv Eikon. The period that will be examined for this study is April 2015 to March 2025. After filtering for the relevant categories and the time period covered by this study, a total of 37 funds were collected, of which 25 were actively managed and 12 were passively managed. Although some objective information was available from Refinitiv Eikon, the data was supplemented with variables from a different source, Swedish House of Finance (Sodini, Fu, and Aytug, 2022). The data from the Swedish House of Finance was used to extract the variables SMB and HML to be able to compute the three-factor model for the funds. The only available data for such variables for the Swedish market was available only to December 2019, which lead to only 56 months being available for evaluation for the three-factor model, compared to 120 months in the rest of the study.

The focus on Swedish equity funds over a ten-year period reflects an effort to ensure consistency in the dataset and enable robust comparisons between management styles. Equity funds are particularly suitable for this type of performance analysis due to their risk-return characteristics and the availability of both active and passive variants. Moreover, the long time horizon helps mitigate short-term market fluctuations and ensures that all included funds have a full performance history over the studied period.

A key limitation of this study relates to survivorship bias, which arises from only including funds that are still active today and have a complete ten-year history. While this selection enables long-term performance comparisons, it risks distorting the results by excluding funds that were liquidated or merged due to poor performance (Shermer, 2014). This bias is particularly relevant for actively managed funds, which are more likely to be closed if they underperform. As a result, the average performance of active funds may appear inflated, since only the more successful ones remain in the sample. In contrast, passive funds that only mirror an index are less likely to be discontinued, making their performance easier to evaluate over time (Elton, Gruber, and Blake, 1996). Unless this bias is explicitly addressed, it may lead to the misleading conclusion that active funds outperform, when in fact many underperforming funds have simply disappeared from view. According to Morningstar, an average of 26 Swedish-registered funds were closed annually over the past decade, including 37 closures in the first eight months of 2019 and 49 in the autumn of 2018 (Morningstar, 2019). Therefore, results from this study must be drawn with a bit of caution, since the survivorship could introduce bias to the results, mainly by favoring the active funds.

To evaluate and compare the performance of active and passive funds, several financial measurements were applied throughout the study. These measurements were employed in order to capture different aspects of the phenomenon under investigation. Each type of measurement offers distinct strengths and serves a specific purpose. By integrating multiple types of measurements, the analysis can address the research questions from several perspectives, thereby increasing the validity and robustness of the results. This multi-measurement approach allowed for a more comprehensive assessment, minimizes potential biases associated with relying on a single method, and enhances the interpretability of the findings.

The total return was used as one way to measure and compare the performance of funds, which uses the following formula:

$$\text{TR } (\%) = \left(\frac{P_E - P_0}{P_0} \right) \times 100 \tag{11}$$

where:

$$P_0 = \text{ is the start value,}$$

$$P_E = \text{ is the ending value.}$$

Total return was used to measure the overall percentage gain of a fund over the specified time period. This metric provides a straightforward way to compare how much a fund has grown from the beginning to the end of the period under study. However, this measure does not account for the risk taken to achieve that return. To evaluate if the risk-adjusted performance is better for active funds, the Sharpe ratio was used. Sharpe ratio uses the following formula:

$$\text{SR} = \frac{R_p - R_f}{\sigma_p} \tag{12}$$

where:

$$R_p = \text{Return of portfolio,}$$

$$R_f = \text{Risk-free interest rate,}$$

$$\sigma_p = \text{standard deviation of portfolio returns.}$$

The Sharpe ratio measures the risk-adjusted return of an investment. It simply measures how much more return you get for each unit of risk. Investors will demand a higher return to accept higher portfolio volatility. Sharpe ratio is a commonly used statistic to rank portfolios in terms of this risk-return trade-off (Bodie, Kane, and Marcus, 2013).

The study also uses the performance measure Jensen's alpha, which was introduced by Jensen (1968). This was used as a performance comparison since it allowed for evaluation of the fund's return above what can be explained by the fund's exposure to market fluctuations (Bodie, Kane, and Marcus, 2013). If $\alpha_i > 0$ it means that the fund delivered returns above the excess return predicted by CAPM. Jensen's alpha was extracted from both the single-factor model in Equation 4 and the three-factor model in Equation 5.

Before proceeding to the performance evaluation and hypothesis testing, descriptive statistics are presented to provide an overview of the selected funds. This includes time series of return for both active and passive funds, scatterplots between the return and fees, and boxplots of the Sharpe ratio. These initial summaries help contextualize the empirical analysis by highlighting basic characteristics and differences between the fund types.

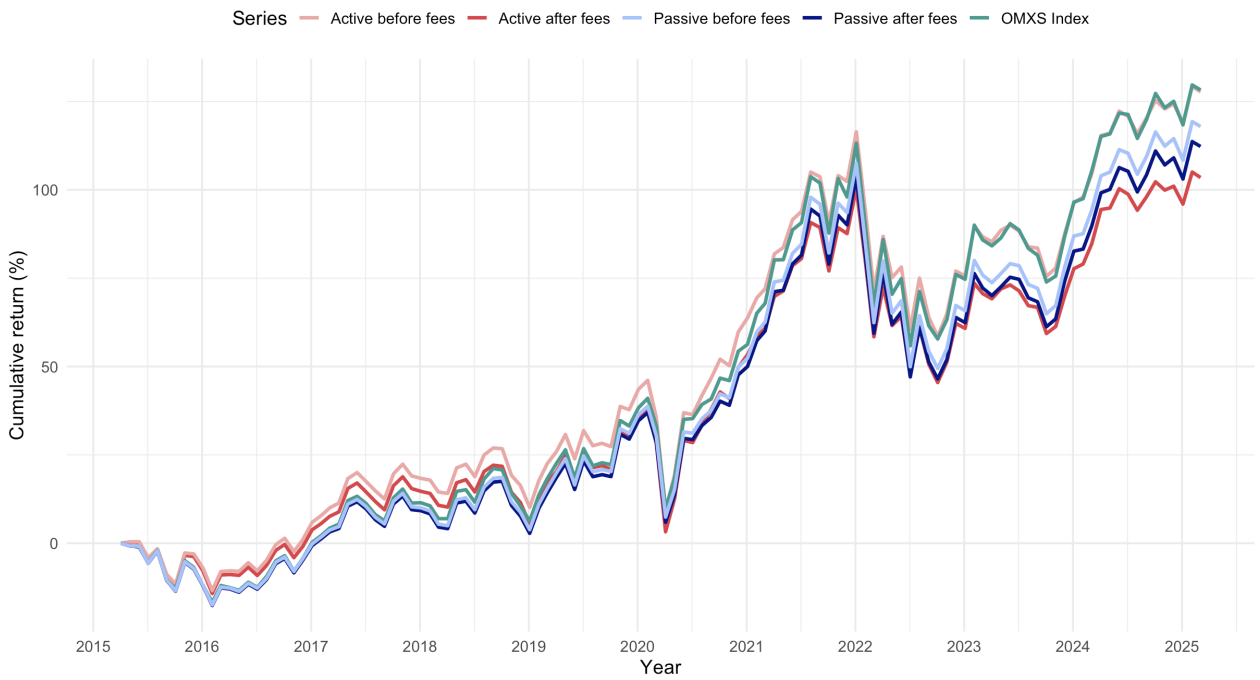


Figure 1: Return before and after fees

Figure 1 presents the cumulative returns over the examined time period for both active and passive funds, shown as gross (before fees) and net returns (after fees). The OMXS index was included as a benchmark to provide context for evaluating the relative performance of the fund types. Looking at the lines for gross return, both fund types generally follow the index closely. Active funds outperform passive funds during several periods, particularly in the years following market dips, such as around 2020 and 2022. This pattern might suggest that active fund managers may have been able to capitalize on market volatility more effectively than passive funds during these times. By the end of the period, active funds managed to perform a mean total return of 127.7%, while the passive funds managed to perform a total return of 117.9% in gross return. This corresponds to a performance gap of 9.79 percentage points over ten years, which represents around 1 percentage point annually.

Looking at the lines representing net return, a substantial difference can be noticed compared to the previous figure. The gaps between active and passive funds are quite smaller compared to before fees, and in many places, the differences are barely noticeable. Both fund types underperform OMXS slightly, indicating that fees may erode some of the market return, especially in active funds. When adjusting for fees, active funds perform worse or the same as the passive funds during major drawdowns in the market. After ten years, passive funds even manage to perform a higher return than active funds, when adjusting for fees. Active funds manage to perform a total return of 103.5% , while passive funds manage to perform a total return of 112.2%. These patterns underscore the importance of considering fees when evaluating fund performance. To gain further insight into how the relative performance of active and passive funds has developed over time, a direct comparison of the fund's return differences throughout the period was evaluated.

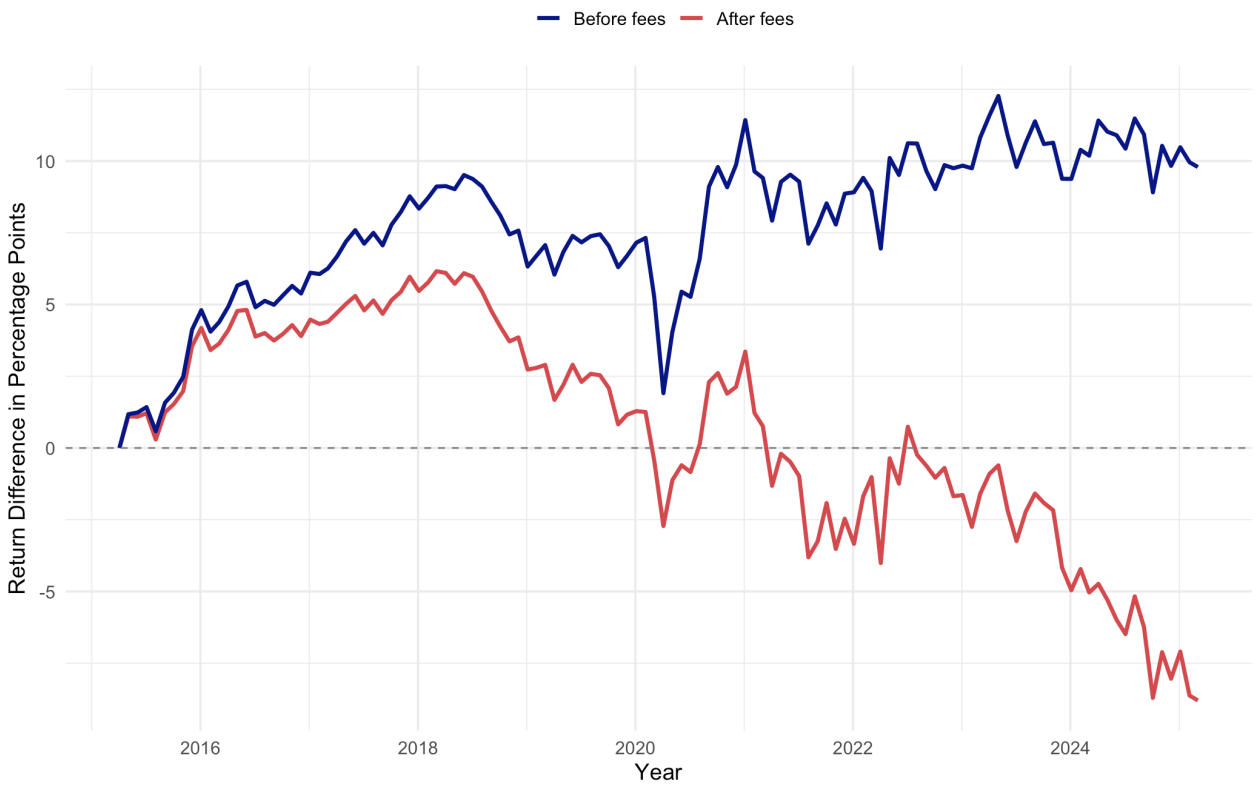


Figure 2: Return Difference Before and After Fees (Active - Passive)

Figure 2 plots the difference in returns between active and passive funds before and after fees. The difference before fees steadily increased from 2015 to 2018. After 2018, the difference fluctuated and changed direction. A sharp dip occurred around 2020, likely caused by the Covid-19 shock, but then recovers quite fast before eventually reaching a difference of 9.79 percentage points between active and passive funds on the final recorded date.

The other line shows the difference between active and passive funds after fees. In contrast to the graph before fees, the graph now looks almost inverted and follows a steady downward trend. The turning point was around 2018-2019, which marks the beginning of the downward trend. The difference eventually reached -8.8 percentage points, indicating that the chosen passive funds even manage to beat the chosen active funds, which was also seen in Figure 1.

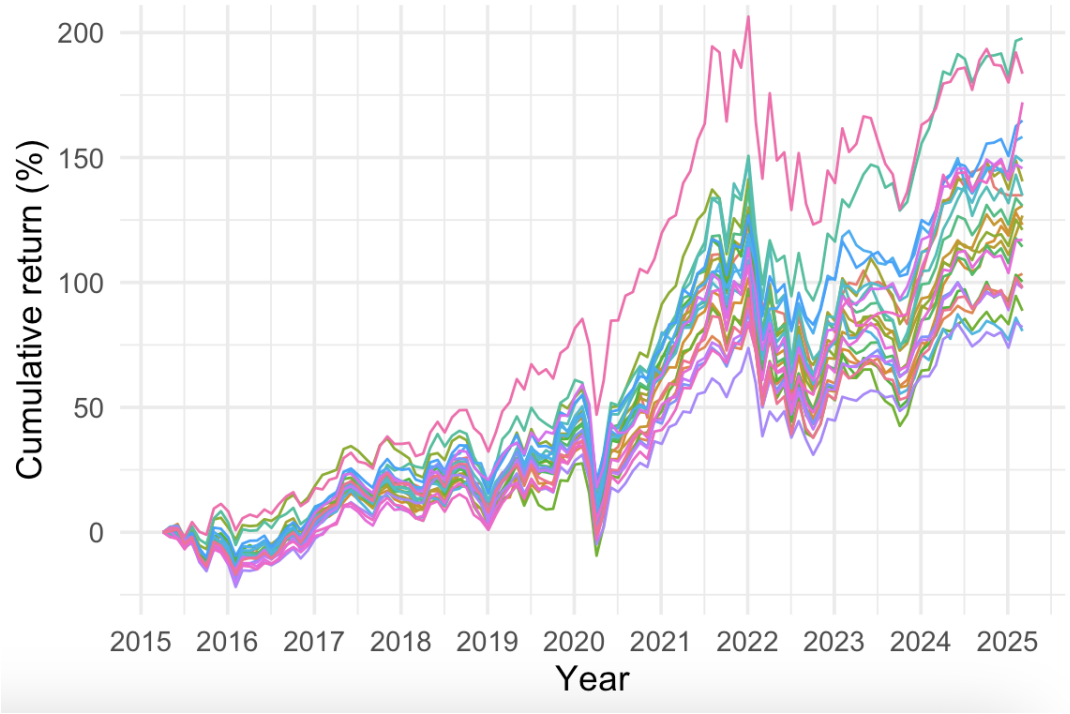


Figure 3: Cumulative returns across active funds (before fees)

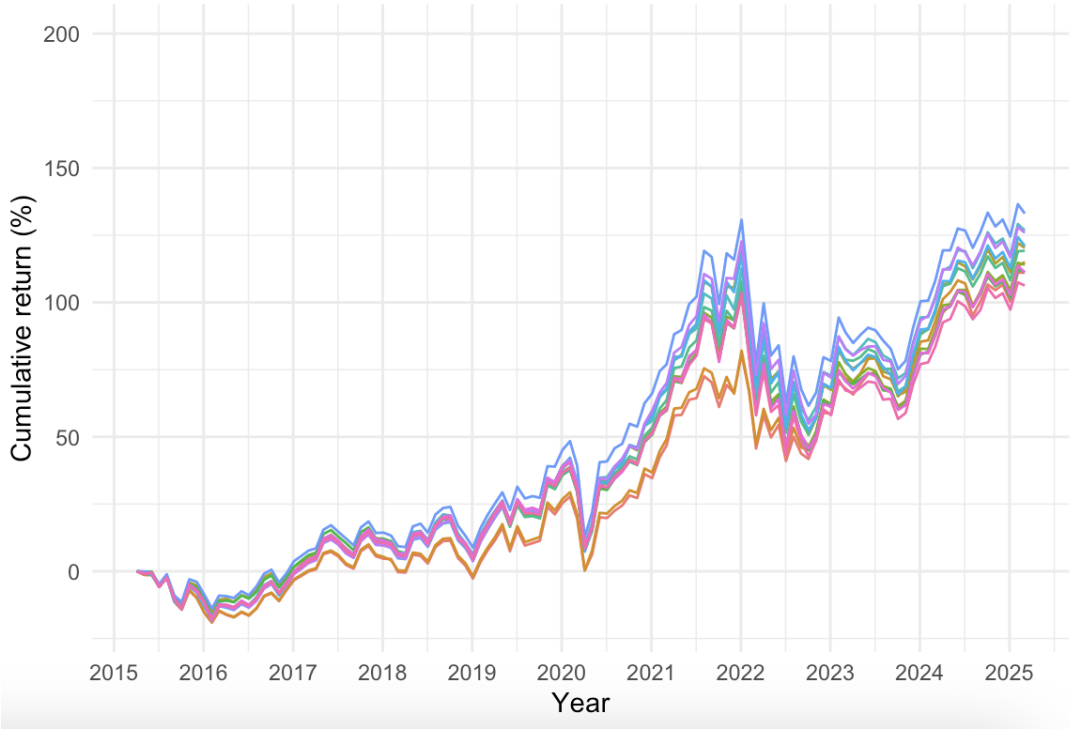


Figure 4: Cumulative returns across passive funds (before fees)

To get a perception of individual performances between fund types, the following Figures 3 and 4 have been displayed. Figure 3 shows the cumulative return for all the active funds in gross return. The graph shows a quite wide spread across the funds, indicating that some have consistently outperformed or underperformed relative to the group.

The best-performing active fund was Handelsbanken Sverige Selektiv, which managed to perform a total return of 197,7%, closely followed by Spiltan Aktiefond Stabil, which managed to perform a total return of 183,6%. Spiltan performed the best throughout most of the 10 years, although it was eventually overtaken toward the end. It is, however, important to note that these funds had high volatility, especially Handelsbanken Sverige Selektiv. The worst-performing fund was Lannebo Sverige Hållbar, which performed a total return of 80.5%. It was tightly followed by Nordea Ideell Aktiefond, which had a total return of 82,1%. The difference between the best-performing active funds and the worst-performing one is around 100 percentage points, further showing the difference in risk and return across active funds. It is important to note that the cumulative returns shown are before fees, and actual returns may therefore be lower depending on the fund's fee structure.

Figure 4 shows the cumulative return across the chosen passive funds. As expected, the graph shows a noticeably narrower spread of lines compared to the spread observed among the active funds, further strengthening the fact that passive funds have less risk and also less difference in return since they mirror an index; there is also little to no active decision-making involved in the portfolio. The best-performing passive fund was SEB Sverige Indexnära, which performed a total return of 133,1%. The worst-performing passive fund was Swedbank Robur Access, which performed a total return of 106,3%. The difference between the best- and worst-performing passive fund is around 26 percentile points, compared to the difference between active funds, which was 100 percentage points.

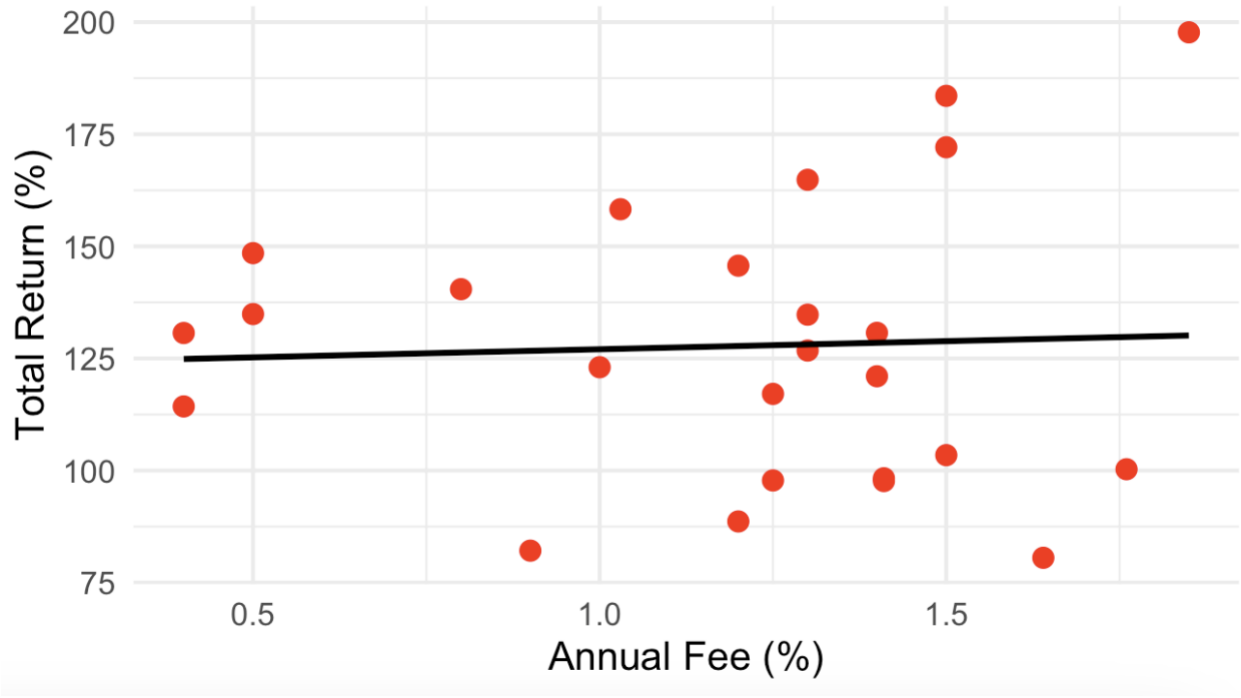


Figure 5: Fee & Total Return Scatterplot: Active funds

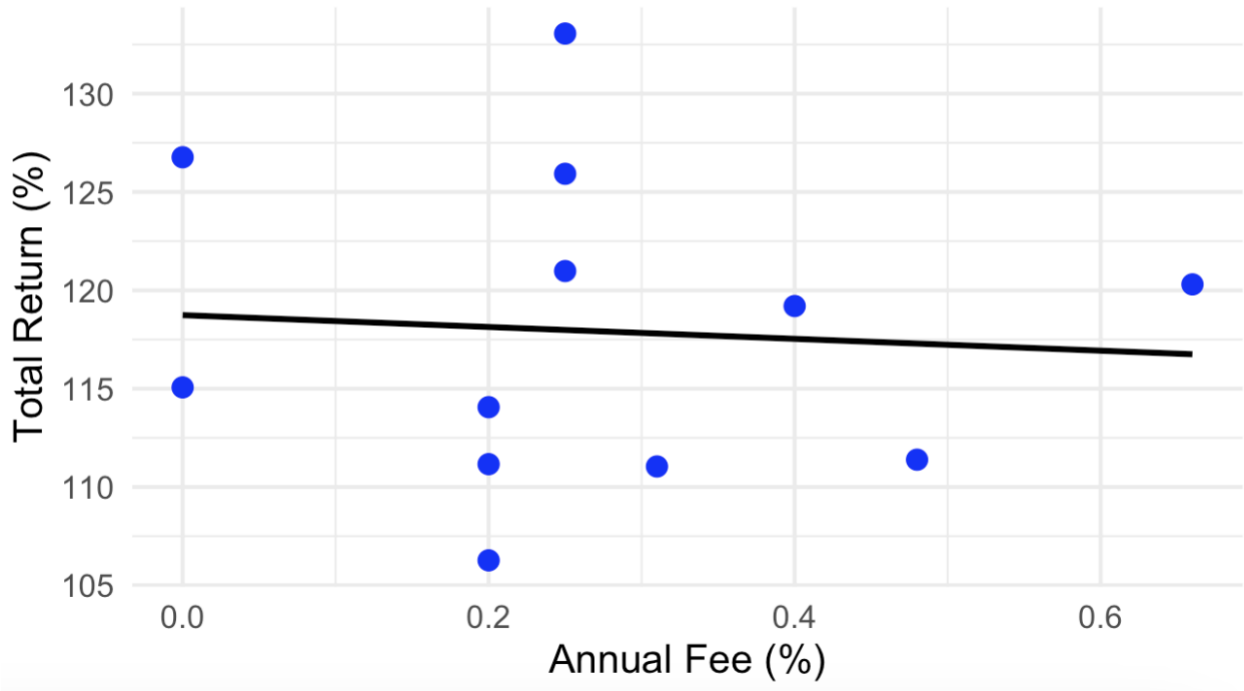


Figure 6: Fee & Total Return Scatterplot: Passive funds

While the selected funds exhibited considerable variation in total return, particularly among the actively managed funds, no clear pattern emerged between fund fees and performance. To examine this further, the relationship between total return and annual management fees was illustrated using scatterplots for both fund types, with a regression line included in each plot. Figure 5 displays the relationship between the annual fee and the total return over the ten-year period for the active funds. Total return was calculated before fees to isolate the fund's gross performance and enable a comparison across different fee levels. The scatterplot shows no visible association between the two variables, and the estimated Pearson correlation coefficient was 0.047. Coefficients and p-values from the regression between gross return for all funds and annual fee are shown in the Appendix, Table 17.

Looking at the funds and their fees, Handelsbanken Sverige Selektiv, which was the top-performing active fund, also had the highest fee. The second-best fund, Spiltan, which managed to get almost the same total return, had the 4th highest fee of 1.5%. Even with the 3rd highest fee of 1.64%, Lannebo Sverige Hållbar had the worst return at the end of the period. Nordea Ideell Aktiefond, which had a similar return, had one of the lowest fees of the active funds, the 20th highest, of 0.9%. Similarly, Figure 6 presents the same relationship for the passive funds, with total return also measured before fees, no clear relationship was observed, and the estimated correlation was -0.071. SEB Sverige Indexnära, which was the best passive fund, had the 5th highest fees across the passive funds of 0.25%. Swedbank Robur Access, which was the worst-performing passive fund, had a lower fee, the 6th highest across passive funds, of 0.2%. There was no clear link between fund performance and fees for either active or passive funds.

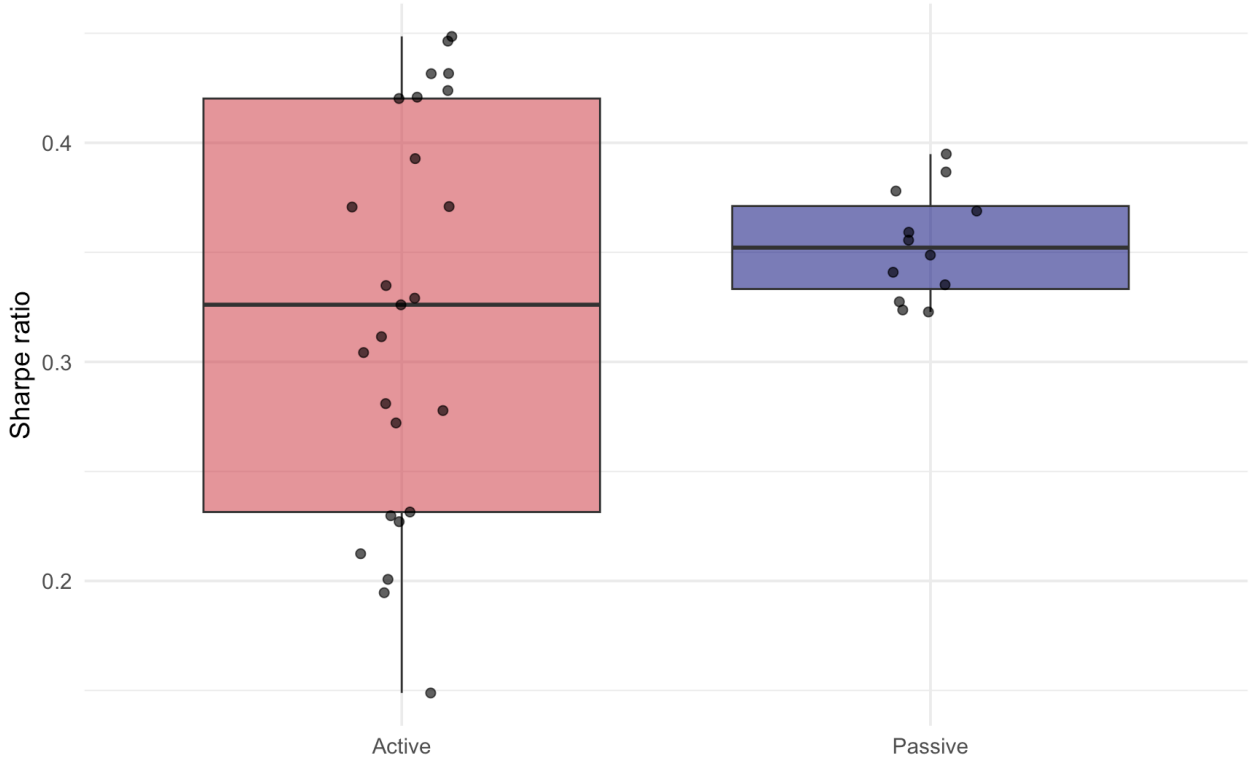


Figure 7: Sharpe ratio boxplot

Finally, Figure 7 displays the distribution of Sharpe ratios for active and passive funds after fees. The plot shows that the active funds exhibited a wider spread, indicating greater variation in risk-adjusted performance within this group. While some active funds achieved relatively high Sharpe ratios, the overall central tendency favored passive funds. The median and mean Sharpe ratios for passive funds were both approximately 0.35, slightly higher than those of active funds, which show a median of around 0.33 and a mean of 0.32. The highest Sharpe ratio among the active funds was 0.45, observed in SEB Swedish Value and Indecap Guide Sverige. Among the passive funds, the top performers, Nordnet Sverige Index and Avanza Zero, both reached a Sharpe ratio of 0.39.

5 Results

This section presents the empirical results observed from the normality and t-tests, single- and three-factor model and finally the bootstrap evaluation.

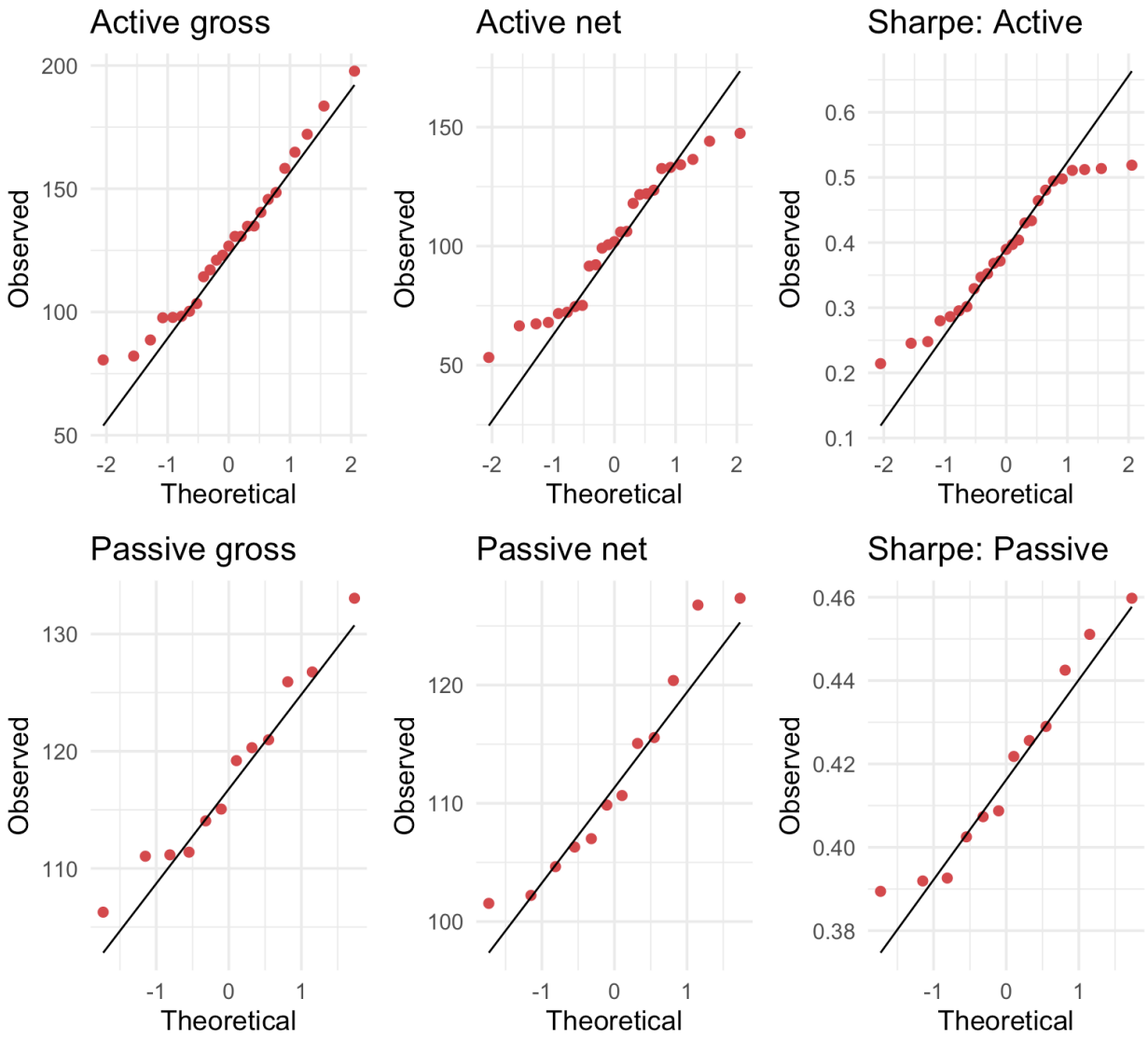


Figure 8: QQ-plots for sample data

As displayed in Figure 8, the observations line up quite well along the diagonal line in the central range. There were some discrepancies from the theoretical line, which might indicate some skewness in the data. The observed deviations from the theoretical normal line in several of the QQ-plots indicate potential departures from normality, particularly in the tails. To formally assess the normality assumption, the Shapiro–Wilk test was also applied to the relevant variables.

Table 1: Results from the Shapiro Wilk test for normally distributed data

Variables	**W-statistic**	**p-value**
Mean total return Active funds gross return	0.9663	0.5542
Mean total return Passive funds gross return	0.9588	0.7669
Mean total return Active funds net return	0.9418	0.1625
Mean total return Passive funds net return	0.9177	0.2674
Sharpe ratio Active funds	0.9379	0.1315
Sharpe ratio Passive funds	0.9416	0.5193

In Table 1 the following W-statistic and p-value are presented. With the null hypothesis that the data is normal, the p-value was too high to reject the null hypothesis for all of the variables and for both groups at the significance level 5%. This indicates that the assumption of normally distributed data holds sufficiently well for all variables, and parametric tests such as the t-test can be appropriately applied.

Table 2: Welch Two Sample t-test (H_1: active > passive)

Comparison	**Mean (Active / Passive)**	**df**	***t*-value**	***p*-value**
Gross return	127.73 / 117.93	29.62	1.47	0.076
Net return	102.37 / 112.28	31.99	-1.60	0.941
Sharpe ratio	0.322 / 0.353	30.33	-1.63	0.943

Table 2 presents the results from Welch's two-sample t-tests comparing active and passive funds across three key performance metrics: gross return, net return, and Sharpe ratio. For gross returns, active funds had a higher mean (127.73%) compared to passive funds (117.93%). However, the observed t-value of 1.47 corresponded to a p-value of 0.076, which was above the 5% significance level; this means that the difference is not statistically significant. In terms of net returns, passive funds actually outperformed on average (112.28%) than active funds (102.37%). The negative t-value of -1.60 and the p-value of 0.941 indicate no support for the hypothesis that active funds performed better after fees.

The Sharpe ratio comparison showed similar results. Active funds achieved a lower average risk-adjusted return (0.322) compared to passive funds (0.353), and the test yields a p-value of 0.943. This result suggests that active funds do not have a higher Sharpe ratio than passive funds at the significance level 5%. Overall, none of the tests provide statistically significant evidence that active funds outperform passive funds in terms of gross return, net return, or Sharpe ratio.

Table 3: Alpha – Summary Statistics (Single-Factor)

α	**Active funds**	**Passive funds**
Mean (gross)	-0.055	-0.098
Std. deviation (gross)	0.118	0.027
Mean (net)	-0.155	-0.108
Std. deviation (net)	0.123	0.03

Table 4: Alpha – Summary Statistics (Three-Factor, 2015-2019)

α	**Active funds**	**Passive funds**
Mean (gross)	-0.013	-0.168
Std. deviation (gross)	0.196	0.08
Mean (net)	-0.11	-0.183
Std. deviation (net)	0.191	0.083

Tables 3 and 4 summarize the estimated alpha values for active and passive funds under the single-factor and three-factor models. Under the single-factor model, the average gross alpha for active funds was -0.055, while passive funds showed a slightly lower value of -0.098. After adjusting for fees, the mean alpha for active funds drops further to -0.155, whereas the passive funds had a net alpha of -0.108. The standard deviations indicate greater variability among active funds in both gross and net returns.

In the three-factor model, which covers a shorter period (2015–2019), the gross alpha of active funds improved slightly to -0.013, while passive funds exhibited a more negative alpha of -0.168. After accounting for fees, active funds had a mean net alpha of -0.11, compared to -0.18 for passive funds. Again, standard deviations were higher among active funds, reflecting more dispersed performance outcomes. These results show that, on average, both fund types underperform the market in both the single-factor and three-factor models, as reflected by negative alpha values.

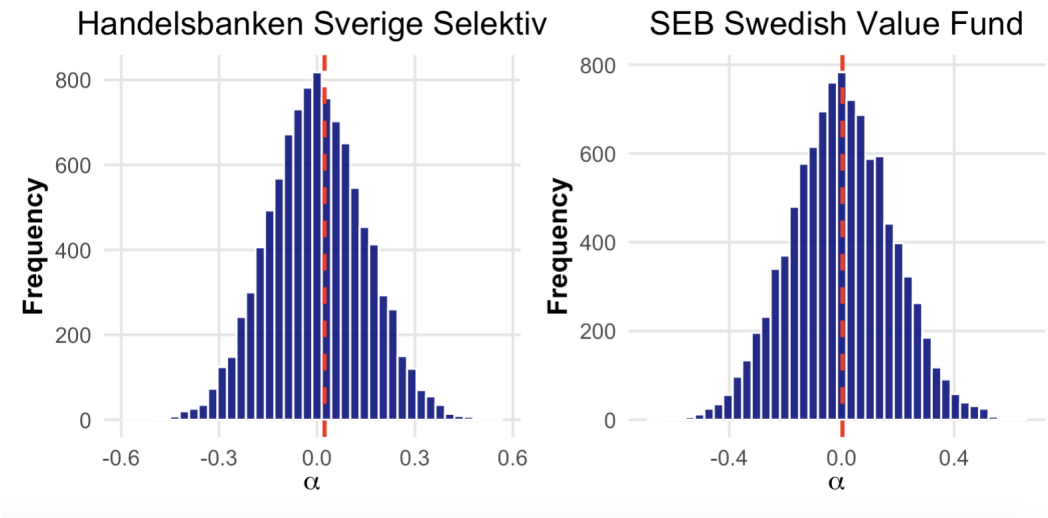


Figure 9: Bootstrap distributions of alphas

Table 5: Bootstrap p-values for top-performing funds (Single-factor, H_1: $\alpha_i > 0$)

Fund	**Bootstrap p-value**
Handelsbanken Sverige Selektiv	0.433
SEB Swedish Value Fund	0.493

A bootstrap for all funds, both from the single- and three-factor model, was performed with the p-values shown in the Appendix, Table 15 and 16. The two funds with the highest net alphas from the single- and three-factor model were investigated further.

In Figure 9 the bootstrap result from Handelsbanken Sverige Selektiv and SEB Swedish Value fund is illustrated. Each histogram of the 10,000 replicates of alphas under the null hypothesis imposing $\alpha_i = 0$ based on the single-factor model is presented for the top two performing (alpha) active funds in the sample, where alpha is based on net returns. The actual alpha for each fund is represented by a red dashed line. Visually, the observed alphas lie close to the center of the bootstrap distributions, suggesting that the values could reasonably occur under the null hypothesis. The estimated p-values are represented below in Table 5. For both Handelsbanken Sverige Selektiv and SEB Swedish Value Fund, the p-values (0.433 and 0.493) were too high to reject the null hypothesis that alpha is less than or equal to zero at the significance level 5%.

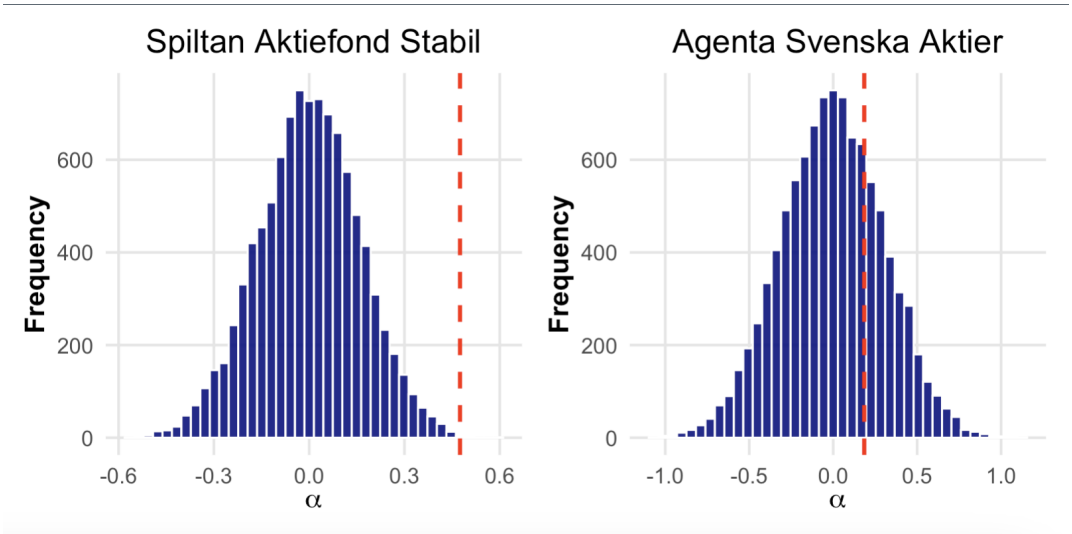


Figure 10: Bootstrap distributions of alphas

Table 6: Bootstrap p-values for top-performing funds (Three-factor, H_1: $\alpha_i > 0$, 2015-2019)

Fund	**Bootstrap p-value**
Spiltan Aktiefond Stabil	0.001
Agenta Svenska Aktier	0.273

Figure 10 shows the bootstrap result from Spiltan Aktiefond Stabil and Agenta Svenska Aktier. The figure similarly displays the distribution of the 10,000 bootstrap replicates under the null hypothesis imposing $\alpha_i = 0$, but for the top two performing (alpha) active funds based on the three-factor model. The red dashed line represents the actual net alpha for each fund, just like in Figure 9. Visually, the alpha for Spiltan Aktiefond Stabil is in the far right tail of the bootstrap distribution. For Agenta Svenska Aktier, the actual alpha is on the right side of the distribution but not too far off the center, indicating it could reasonably occur under the null hypothesis. The p-value for Spiltan Aktiefond Stabil was 0.001, which means that the null hypothesis that alpha is less than or equal to zero was rejected in favor of the alternative that alpha is greater than zero at the significance level 5% for the three-factor model. For Agenta Svenska Aktier, the estimated p-value was 0.273, which means the null hypothesis that alpha is less than or equal to zero was not rejected at the significance level 5%. Since only Spiltan showed a significant p-value across both passive and active funds, no additional funds were discussed further.

6 Summary/Discussion

The goal of the report has been to study whether active funds offer added value compared to passive funds. The results indicate that active funds do not, on average, perform better than passive funds after adjusting for fund fees.

A pattern that emerged both from plotting the data and also analyzing the alphas between active and passive funds from the regression models was that active funds showed greater variation in performance than passive funds. Passive funds delivered more stable, predictable results, both in total return, Sharpe ratio, and alphas. This goes in line with what is generally true for passive funds: they follow an index and usually exhibit less risk than active funds. The results also did not find any meaningful correlation between fund fees and performance, suggesting that the higher fees could not deliver any added value. For both net return and Sharpe ratio, passive funds outperformed active funds on average.

The two-sample t-test for both total return and Sharpe ratio resulted in p-values that were too high to reject the null hypothesis that active funds had a mean equal to or worse than passive funds. Furthermore, the alpha values from the single- and three-factor regression models showed that while many active funds do generate high alphas before fees, most funds do not generate excess return compared to the market index after fees. This suggests that even if some funds succeed with positive alphas, most funds lose this benefit when accounting for fund fees.

Studying the two funds with the highest alphas from the single- and three-factor model with the bootstrap methodology showed that only one fund, Spiltan Akitefond Stabil, had a statistically significant alpha, indicating that only this fund might be able to pick stocks well enough to more than cover their fund fees, while controlling for luck. This also goes in line with what (Kosowski et al., 2006) found, with only a minority of their chosen 4200 funds exhibiting significant values. This suggests that even among the best-performing funds, finding funds that manage to consistently outperform the market is extremely hard, a result that also goes in line with the Efficient Market Hypothesis by (Fama, 1970). Important to note is that Spiltan Aktiefond Stabil from the three-factor model used less data, from 2015 to 2019, compared to the rest of the analysis from 2015 to 2025, which might affect the reliability of the results.

The results from the study are also in line with previous research, such as the study by (Kosowski et al., 2006) and (Fama and French, 2010) which showed that active funds are not, on average, compensated by superior performance when adding fund fees. While the results show that the Swedish passive funds outperform actively managed ones on average after fees, certain active funds do achieve a stronger return than passive funds. However, just like (Kosowski et al., 2006) showed in their results, these active funds represent a minority, motivating that in the long term, passive funds would be a more favorable choice for most investors. Furthermore, the fact that only one fund in the sample had a significant alpha while controlling for luck highlights the difficulty in finding the right active fund in advance.

These results have important implications for Swedish investors. Identifying the right active fund in advance is difficult, especially for non-professional investors, particularly since only one fund showed statistically significant evidence of skill sufficient to more than cover fees while controlling for luck. It all comes down to the amount of risk an investor is willing to take. Also important to note is that no matter the results of this report, historical returns are not a guarantee of future results.

It should also be noted that the choice of period may have influenced the results of the report, and it cannot be guaranteed that the results are generalizable for all periods. Furthermore, since this study focuses on the Swedish fund market, the results may not apply to other countries and their markets. Given that this study is limited to the Swedish market, an interesting focus for further research would be to test whether these results hold if tested simultaneously across several international markets. It would also be interesting to study a longer period than ten years. The problem with this is that the longer the time period becomes, the harder it becomes to find active funds that have been active for the whole period. This also comes with the problem of survivorship bias, with funds still being active for the chosen period typically being the ones that still are successful, while the funds not performing well are the ones shut down, making analysis very hard. This issue is also relevant in this study; however, since the results still indicate that active funds do not outperform passive funds after fees, including the closed active funds that performed worse, would likely only strengthen this result.

The results from this study indicate that while active funds on average beat passive funds before fees, there is little significant evidence that they do so on average when accounting for fees. Although one fund exhibited a significant positive alpha from the bootstrap, indicating that this fund may choose stocks to more than cover the fund fees, the results show that these instances are very rare. Taking this into account, these findings do not support active funds consistently adding value, making passive funds a more favorable choice for most investors.

References

Bodie, Z., A. Kane, and A. J. Marcus (2013). *Essentials of Investments*. 9th ed. New York: McGraw-Hill/Irwin.

Cryer, J. D. and K. Chan (2008). *Time Series Analysis: With Applications in R*. 2nd ed. New York: Springer.

De Bondt, Werner F. M. and Richard Thaler (1985). "Does the Stock Market Overreact?" In: *The Journal of Finance* 40.3. Online, pp. 793–805. DOI: 10.2307/2327804. URL: https://doi.org/10.2307/2327804 (visited on 05/20/2025).

Devore, J. L. and K. N. Berk (2012). *Modern Mathematical Statistics with Applications*. 2nd ed. New York: Springer.

Elton, E. J., M. J. Gruber, and C. R. Blake (1996). "Survivor Bias and Mutual Fund Performance". In: *The Review of Financial Studies* 9.4. Online, pp. 1097–1120. URL: https://academic.oup.com/rfs/article-abstract/9/4/1097/1580100 (visited on 05/20/2025).

Fama, Eugene F. (1970). "Efficient Capital Markets: A Review of Theory and Empirical Work". In: *The Journal of Finance* 25.2. Online, pp. 383–417. DOI: 10.2307/2325486. URL: https://doi.org/10.2307/2325486 (visited on 05/20/2025).

Fama, Eugene F. and Kenneth R. French (1992). "The Cross-Section of Expected Stock Returns". In: *The Journal of Finance* 47.2. Online, pp. 427–465. DOI: 10.1111/j.1540-6261.1992.tb04398.x. URL: https://doi.org/10.1111/j.1540-6261.1992.tb04398.x (visited on 05/20/2025).

— (2010). "Luck versus Skill in the Cross-Section of Mutual Fund Returns". In: *The Journal of Finance* 65.5, pp. 1915–1947. DOI: 10.1111/j.1540-6261.2010.01598.x. URL: https://doi.org/10.1111/j.1540-6261.2010.01598.x (visited on 05/20/2025).

Fondbolagens förening (2025). *Rekordår för fondsparandet 2024*. Online. URL: https://www.fondbolagen.se/aktuellt/pressrum/pressmeddelanden/rekordar-for-fondsparandet-2024 (visited on 05/20/2025).

— (n.d.). *Så blev Sverige världsbäst på fondsparande*. Online. URL: https://www.fondbolagen.se/fakta_index/sa-blev-sverige-varldsledande-pa-fondsparande (visited on 05/20/2025).

Handelsbanken (n.d.). *Vad är fonder och hur fungerar det egentligen?* Online. URL: https://www.handelsbanken.se/sv/ekonomi-i-livet/privatekonomi/spara-klokt/fondhjalpen/vad-ar-en-fond (visited on 05/20/2025).

JMP (2024). *The t-Test*. Online. URL: https://www.jmp.com/en/statistics-knowledge-portal/t-test (visited on 05/20/2025).

Kosowski, Robert et al. (2006). "Can Mutual Fund 'Stars' Really Pick Stocks? New Evidence from a Bootstrap Analysis". In: *The Journal of Finance* 61.6. Online, pp. 2551–2595. DOI: 10.1111/j.1540-6261.2006.01013.x. URL: https://doi.org/10.1111/j.1540-6261.2006.01013.x (visited on 05/20/2025).

Morningstar (2019). *Svensk registrerade fonder som har lagts ner*. Online. URL: https://www.morningstar.se/se/news/197822/halvering-av-antalet-nya-fonder.aspx (visited on 05/20/2025).

Roll, Richard (1977). "A Critique of the Asset Pricing Theory's Tests Part I: On Past and Potential Testability of the Theory". In: *Journal of Financial Economics* 4.2. Online, pp. 129–176. DOI: 10.1016/0304-405X(77)90009-5. URL: https://doi.org/10.1016/0304-405X(77)90009-5 (visited on 05/20/2025).

Shapiro, S. S. and M. B. Wilk (1965). "An analysis of variance test for normality (complete samples)". In: *Biometrika* 52.3–4. Online, pp. 591–611. URL: https://doi.org/10.2307/2333709 (visited on 05/20/2025).

Sharpe, William F. (1964). "Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk". In: *The Journal of Finance* 19.3. Online, pp. 425–442. DOI: 10.1111/j.1540-6261.1964.tb02865.x. URL: https://doi.org/10.1111/j.1540-6261.1964.tb02865.x (visited on 05/20/2025).

Shermer, M. (2014). *How the Survivor Bias Distorts Reality*. Online. URL: https://www.scientificamerican.com/article/how-the-survivor-bias-distorts-reality/ (visited on 05/20/2025).

Shiller, R. J. (1981). *Do Stock Prices Move Too Much to be Justified by Subsequent Changes in Dividends?* Tech. rep. Online. Cambridge, Mass: National Bureau of Economic Research. URL: https://uub.primo.exlibrisgroup.com/discovery/fulldisplay?context=L&vid=46LIBRIS_UUB:UUB&search_scope=MyInst_and_CI&tab=Everything&docid=alma991018478408607596 (visited on 05/20/2025).

Söderberg&Partners (2014). *Passivt eller aktivt förvaltade fonder – vad ska jag välja?* Online. URL: https://www.soderbergpartners.se/insikt/artiklar/finans/passivt-eller-aktivt-forvaltade-fonder--vad-ska-jag-valja/ (visited on 05/20/2025).

Sodini, Paolo, Yao Fu, and Huseyin Aytug (2022). *Construction of the Fama-French-Carhart Four Factors Model for the Swedish Stock Market Using the Finbas Data*. Swedish House of Finance Research Data Center. Online. URL: https://www.hhs.se/en/houseoffinance/data-center/fama-french-factors/ (visited on 05/20/2025).

Triola, M. F (2015). *Selected chapters from Elementary Statistics*. 12th ed. Harlow: Pearson Education Limited.

Tuvhag, E. (2023). *Svenskar bäst på att spara i hela Europa*. Online. URL: https://www.svd.se/a/GpVQ/svenskar-bast-pa-att-spara-i-hela-europa (visited on 03/31/2025).

Appendix

Table 7: Active funds

Name	Gross Tot Return (%)	Net Tot Return (%)	Annual fee (%)
Agenta Svenska Aktier	134.9	123.51	0.5
Aktie-Ansvar Sverige	103.43	75.12	1.5
Carnegie Spin-Off	123.04	101.88	1
Carnegie Sverigefond	130.71	100.6	1.4
Case Högutdelande Bolag	126.76	99.16	1.3
Cliens Sverige	140.45	122.04	0.8
Cliens Sverige Fokus	121.03	92.19	1.4
D&G Aktiefond	88.65	67.37	1.2
Enter Select	100.3	67.96	1.76
Folksam LO Sverige	114.3	105.95	0.4
Folksam LO Västfonden	130.67	121.68	0.4
Handelsbanken Sverige Selektiv	197.72	147.39	1.85
Humle Sverigefond	134.76	106.19	1.3
Indecap Guide Sverige	148.48	136.43	0.5
Lannebo Sverige Hållbar	80.55	53.24	1.64
Lannebo Sverige Plus	158.27	133.07	1.03
Länsförsäkringar Sverige Vision	164.83	132.61	1.3
Nordea Alfa	98.28	72.24	1.41
Nordea Ideell Aktiefond	82.14	66.52	0.9
Nordea Swedish Stars	97.66	71.7	1.41
PriorNilsson Sverige Aktiv	145.68	117.96	1.2
SEB Sverige Expanderad	117.1	91.64	1.25
SEB Swedish Value Fund	172.09	134.22	1.5
Spiltan Aktiefond Stabil	183.55	144.09	1.5
Swedbank Robur Sverige	97.8	74.6	1.25
Mean	127.73	102.37	1.19
Standard deviation	31.31	28.12	0.41

Table 8: Active funds (Annual)

Name	Geometrical mean (%)	Arithmetical mean post fee (%)	Volatility (%)	Sharpe-ratio
Agenta Svenska Aktier	8.38	8.8	15.86	0.42
Aktie-Ansvar Sverige	5.76	6.23	15.08	0.27
Carnegie Spin-Off	7.28	7.78	17.32	0.33
Carnegie Sverigefond	7.21	7.57	16.54	0.33
Case Högutdelande Bolag	7.13	7.58	17.92	0.3
Cliens Sverige	8.3	8.78	17.93	0.37
Cliens Sverige Fokus	6.75	7.38	18.9	0.28
D&G Aktiefond	5.28	5.98	19.18	0.2
Enter Select	5.32	5.93	19.53	0.19
Folksam LO Sverige	7.49	8	17.54	0.33
Folksam LO Västfonden	8.29	8.81	18.03	0.37
Handelsbanken Sverige Selektiv	9.48	9.67	17.48	0.43
Humle Sverigefond	7.5	7.97	18.76	0.31
Indecap Guide Sverige	8.99	9.41	16.3	0.45
Lannebo Sverige Hållbar	4.36	4.79	17.89	0.15
Lannebo Sverige Plus	8.83	9.45	17.28	0.42
Länsförsäkringar Sverige Vision	8.81	9.12	16.21	0.43
Nordea Alfa	5.59	6.01	16.76	0.23
Nordea Ideell Aktiefond	5.23	5.7	16.8	0.21
Nordea Swedish Stars	5.55	6.04	17.02	0.23
PriorNilsson Sverige Aktiv	8.1	8.54	16.31	0.39
SEB Sverige Expanderad	6.72	7.07	17.57	0.28
SEB Swedish Value Fund	8.88	9.42	16.26	0.45
Spiltan Aktiefond Stabil	9.33	9.56	17.68	0.42
Swedbank Robur Sverige	5.73	6.25	18.17	0.23
Mean	7.21	7.67	17.37	0.32
Standard deviation	1.51	1.47	1.08	0.09

Table 9: Passive funds

Name	Gross Tot Return (%)	Net Tot Return (%)	Annual fee (%)
Aktiespararna Topp Sverige	111.04	104.64	0.31
Avanza Zero	115.06	115.06	0
Handelsbanken Sverige Index Criteria	120.3	106.3	0.66
Länsförsäkringar Sverige Index	114.06	109.85	0.2
Nordea Inst Aktie Sverige	111.39	101.54	0.48
Nordea Sverige Passiv	119.20	110.66	0.4
Nordnet Sverige Index	126.77	126.77	0
SEB Sverige Indexfond	120.98	115.56	0.25
SEB Sverige Indexnära	133.06	127.34	0.25
Skandia Sverige Exponering	125.92	120.39	0.25
Storebrand Sverige	111.16	107.01	0.2
Swedbank Robur Access Sverige	106.27	102.21	0.2
Mean	117.93	112.28	0.32
Standard deviation	7.88	8.88	0.15

Table 10: Passive funds (Annual)

Name	Geometrical mean (%)	Arithmetical mean (%)	Volatility (%)	Sharpe-ratio
Aktiespararna Topp Sverige	7.42	7.95	16.2	0.36
Avanza Zero	7.96	8.5	16.47	0.39
Handelsbanken Sverige Index Criteria	7.51	7.96	17.81	0.33
Länsförsäkringar Sverige Index	7.69	8.12	17.59	0.34
Nordea Inst Aktie Sverige	7.26	7.68	17.14	0.32
Nordea Sverige Passiv	7.74	8.2	17.08	0.36
Nordnet Sverige Index	8.53	9	17.4	0.39
SEB Sverige Indexfond	7.98	8.49	18.23	0.35
SEB Sverige Indexnära	8.56	8.99	18.15	0.38
Skandia Sverige Exponering	8.22	8.74	17.93	0.37
Storebrand Sverige	7.55	7.97	17.43	0.34
Swedbank Robur Access Sverige	7.3	7.74	17.37	0.32
Mean	7.81	8.28	17.4	0.35
Standard deviation	0.45	0.46	0.62	0.02

Table 11: Single factor model Active funds

Name	α_{gross}	α_{net}	$\beta_{M,gross}$	$\beta_{M,net}$
Agenta Svenska Aktier	0.025	-0.017	0.75	0.75
Aktie-Ansvar Sverige	-0.13	-0.256	0.809	0.808
Carnegie Spin-Off	-0.054	-0.138	0.897	0.896
Carnegie Sverigefond	-0.037	-0.155	0.895	0.894
Case Högutdelande Bolag	-0.068	-0.177	0.990	0.989
Cliens Sverige	-0.01	-0.078	0.967	0.967
Cliens Sverige Fokus	-0.086	-0.204	1.022	1.021
D&G Aktiefond	-0.218	-0.319	1.029	1.028
Enter Select	-0.172	-0.32	1.058	1.056
Folksam LO Sverige	-0.111	-0.144	0.962	0.962
Folksam LO Västfonden	-0.053	-0.086	0.992	0.992
Handelsbanken Sverige Selektiv	0.18	0.023	0.929	0.927
Humle Sverigefond	-0.038	-0.148	1.023	1.022
Indecap Guide Sverige	0.026	-0.016	0.887	0.887
Lannebo Sverige Hållbar	-0.251	-0.389	0.963	0.962
Lannebo Sverige Plus	0.076	-0.011	0.878	0.877
Länsförsäkringar Sverige Vision	0.079	-0.031	0.883	0.882
Nordea Alfa	-0.156	-0.275	0.884	0.883
Nordea Ideell Aktiefond	-0.227	-0.302	0.882	0.881
Nordea Swedish Stars	-0.161	-0.28	0.9	0.899
PriorNilsson Sverige Aktiv	0.023	-0.078	0.874	0.873
SEB Sverige Expanderad	-0.096	-0.201	0.953	0.952
SEB Swedish Value Fund	0.129	0.003	0.819	0.818
Spiltan Aktiefond Stabil	0.126	-0.001	0.965	0.963
Swedbank Robur Sverige	-0.177	-0.282	0.986	0.985
Mean	-0.055	-0.155	0.928	0.927
Standard deviation	0.118	0.123	0.075	0.075

Table 13: Three-factor model Active funds

Name	α_{gross}	α_{net}	$\beta_{M,gross}$	$\beta_{M,net}$	$\beta_{SMB,gross}$	$\beta_{SMB,net}$	$\beta_{HML,gross}$	$\beta_{HML,net}$
Agenta Svenska Aktier	0.227	0.185	0.791	0.79	0.005	0.005	0.04	0.04
Aktie-Ansvar Sverige	-0.133	-0.257	0.929	0.928	0.026	0.026	0.12	0.12
Carnegie Spin-Off	0.029	-0.054	0.818	0.817	0.103	0.103	0.217	0.217
Carnegie Sverigefond	-0.117	-0.234	1.043	1.042	0.052	0.052	0.081	0.081
Case Högutdelande Bolag	-0.019	-0.127	0.992	0.991	0.001	0.001	-0.001	-0.001
Cliens Sverige	-0.105	-0.171	0.997	0.997	0.049	0.049	-0.05	0.05
Cliens Sverige Fokus	0.074	-0.043	1.038	1.036	0.082	0.082	-0.077	0.077
D&G Aktiefond	-0.279	-0.379	1.002	1.001	0.078	0.078	-0.048	-0.048
Enter Select	0.118	0.029	0.976	0.975	-0.046	-0.046	-0.188	-0.188
Folksam LO Sverige	-0.164	-0.197	1.068	1.068	0.026	0.026	-0.014	-0.014
Folksam LO Västfonden	-0.094	-0.127	1.058	1.058	0.007	0.007	-0.005	-0.005
Handelsbanken Sverige Selektiv	0.264	0.109	0.986	0.984	-0.091	-0.091	-0.323	-0.323
Humle Sverigefond	0.086	-0.023	0.938	0.937	-0.018	-0.018	0.046	0.046
Indecap Guide Sverige	-0.023	-0.065	0.999	0.998	0.017	0.017	0.042	0.042
Lannebo Sverige Hållbar	0.021	-0.116	0.956	0.956	-0.124	-0.124	0.021	0.021
Lannebo Sverige Plus	-0.002	-0.088	0.995	0.994	-0.074	-0.074	0.023	0.023
Länsförsäkringar Sverige Vision	0.001	-0.108	1.073	1.071	0.031	0.03	-0.075	-0.075
Nordea Alfa	-0.136	-0.253	1.03	1.03	-0.041	-0.041	-0.019	-0.019
Nordea Ideell Aktiefond	-0.241	-0.316	1.014	1.013	-0.113	-0.113	0.006	0.006
Nordea Swedish Stars	-0.177	-0.294	1.045	1.044	-0.042	-0.042	0.033	0.033
PriorNilsson Sverige Aktiv	0.226	0.126	0.846	0.845	-0.004	-0.004	0.084	0.084
SEB Sverige Expanderad	-0.122	-0.226	0.958	0.957	-0.025	-0.025	-0.053	-0.053
SEB Swedish Value Fund	-0.12	-0.245	0.949	0.948	-0.004	-0.004	-0.03	-0.03
Spiltan Aktiefond Stabil	**0.601**	**0.475**	0.863	0.862	-0.091	-0.091	-0.244	-0.243
Swedbank Robur Sverige	-0.259	-0.362	1.095	1.094	0.02	0.02	0.004	0.004
Mean	-0.013	-0.11	0.978	0.977	-0.007	-0.007	-0.023	-0.006
Standard deviation	0.196	0.191	0.08	0.08	0.06	0.06	0.109	0.112

Table 12: Single factor model Passive funds

Name	α_{gross}	α_{net}	$\beta_{M,gross}$	$\beta_{M,net}$
Aktiespararna Topp Sverige	-0.109	-0.115	0.873	0.873
Avanza Zero	-0.095	-0.095	0.888	0.888
Handelsbanken Sverige Index Criteria	-0.093	-0.135	0.986	0.985
Länsförsäkringar Sverige Index	-0.115	-0.116	0.971	0.97
Nordea Inst Aktie Sverige	-0.123	-0.153	0.947	0.947
Nordea Sverige Passiv	-0.092	-0.11	0.945	0.944
Nordnet Sverige Index	-0.067	-0.067	0.966	0.966
SEB Sverige Indexfond	-0.093	-0.099	1.011	1.011
SEB Sverige Indexnära	-0.049	-0.06	1.009	1.009
Skandia Sverige Exponering	-0.072	-0.076	0.991	0.991
Storebrand Sverige	-0.126	-0.128	0.963	0.963
Swedbank Robur Access Sverige	-0.144	-0.145	0.958	0.958
Mean	-0.098	-0.108	0.959	0.959
Standard deviation	0.027	0.03	0.043	0.043

Table 14: Three-factor model Passive funds

Name	α_{gross}	α_{net}	$\beta_{M,gross}$	$\beta_{M,net}$	$\beta_{SMB,gross}$	$\beta_{SMB,net}$	$\beta_{HML,gross}$	$\beta_{HML,net}$
Aktiespararna Topp Sverige	-0.317	-0.343	1.047	1.045	0.061	0.053	0.04	0.015
Avanza Zero	-0.32	-0.32	1.043	1.043	0.046	0.046	0.058	0.058
Handelsbanken Sverige Index Criteria	-0.116	-0.138	1.003	1.001	0.001	-0.004	0.029	0.013
Länsförsäkringar Sverige Index	-0.164	-0.187	1.05	1.048	-0.005	-0.011	0.013	-0.006
Nordea Inst Aktie Sverige	-0.199	-0.212	1.037	1.035	0.003	-0.001	0.061	0.048
Nordea Sverige Passiv	-0.194	-0.203	1.047	1.045	0.001	-0.005	0.027	0.007
Nordnet Sverige Index	-0.145	-0.145	1.025	1.025	0.001	0.001	0.025	0.025
SEB Sverige Indexfond	-0.095	-0.113	1.001	1	-0.024	-0.029	0.028	0.009
SEB Sverige Indexnära	-0.032	-0.035	1.001	1	-0.002	-0.005	-0.001	-0.012
Skandia Sverige Exponering	-0.112	-0.134	1.02	1.018	-0.028	-0.034	0.014	-0.007
Storebrand Sverige	-0.149	-0.170	1.047	1.045	0.001	-0.004	0.032	0.014
Swedbank Robur Access Sverige	-0.176	-0.201	1.054	1.052	0.005	-0.001	0.05	0.029
Mean	-0.168	-0.183	1.031	1.03	0.005	0.001	0.031	0.016
Standard deviation	0.08	0.083	0.02	0.02	0.024	0.024	0.018	0.02

Table 15: Bootstrap p-values Active funds ($H_1 : \alpha_i > 0$)

Name	Single-factor model	Three-factor model (2015-2019)
Agenta Svenska Aktier	0.528	0.273
Aktie-Ansvar Sverige	0.968	0.942
Carnegie Spin-Off	0.788	0.453
Carnegie Sverigefond	0.916	0.989
Case Högutdelande Bolag	0.996	0.999
Cliens Sverige	0.742	0.869
Cliens Sverige Fokus	0.959	0.603
D&G Aktiefond	0.991	0.992
Enter Select	0.998	0.763
Folksam LO Sverige	0.956	0.987
Folksam LO Västfonden	0.859	0.93
Handelsbanken Sverige Selektiv	0.433	0.299
Humle Sverigefond	0.917	0.576
Indecap Guide Sverige	0.557	0.771
Lannebo Sverige Hållbar	0.992	0.747
Lannebo Sverige Plus	0.519	0.658
Länsförsäkringar Sverige Vision	0.598	0.769
Nordea Alfa	0.955	0.908
Nordea Ideell Aktiefond	0.966	0.951
Nordea Swedish Stars	0.963	0.934
PriorNilsson Sverige Aktiv	0.725	0.288
SEB Sverige Expanderad	0.973	0.966
SEB Swedish Value Fund	0.493	0.973
Spiltan Aktiefond Stabil	0.502	**0.001**
Swedbank Robur Sverige	0.996	0.986

Table 16: Bootstrap p-values Passive funds ($H_1 : \alpha_i > 0$)

Name	Single-factor model	Three-factor model (2015-2019)
Aktiespararna Topp Sverige	0.841	0.994
Avanza Zero	0.464	0.995
Handelsbanken Sverige Index Criteria	0.998	0.973
Länsförsäkringar Sverige Index	0.97	0.984
Nordea Inst Aktie Sverige	0.993	0.999
Nordea Sverige Passiv	0.954	0.995
Nordnet Sverige Index	0.938	0.994
SEB Sverige Indexfond	0.999	0.989
SEB Sverige Indexnära	0.998	0.999
Skandia Sverige Exponering	0.9	0.927
Storebrand Sverige	0.983	0.967
Swedbank Robur Access Sverige	0.984	0.989

$$Y_{GrossReturn_i} = \beta_0 + \beta_1 \cdot X_{Fee_i} + \varepsilon_i$$

	β_0	β_1
Estimate	117.715	7.687
Standard error	8.227	7.864
t-statistic	14.308	0.977
p-value	<0.001	0.335

Table 17: Regression between total return (gross) and annual fee